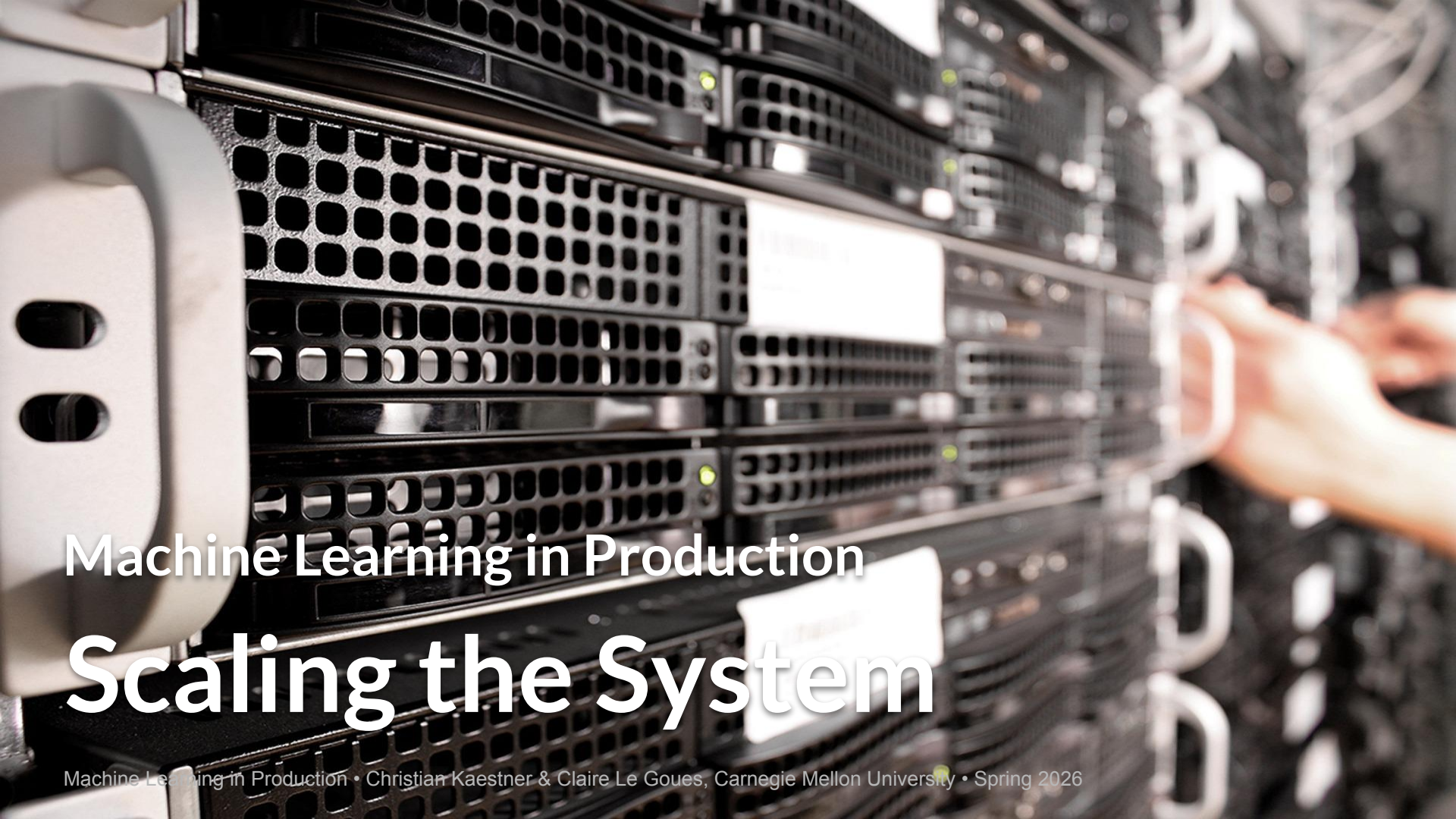


Machine Learning in Production

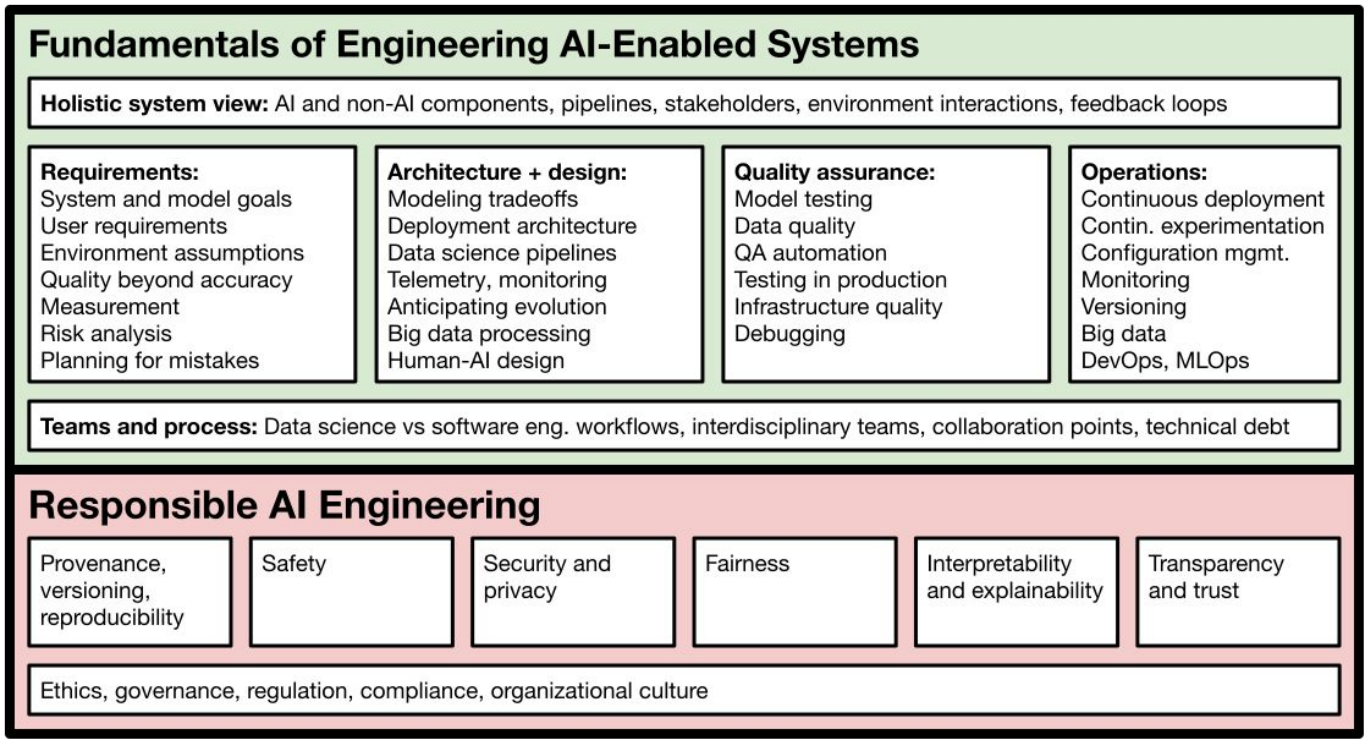
Scaling the System

Administrativa

Last day to opt in (or back out of) participation grading – see Canvas

Midterm here during lecture time on Wednesday

Design and operations



Suggested Readings

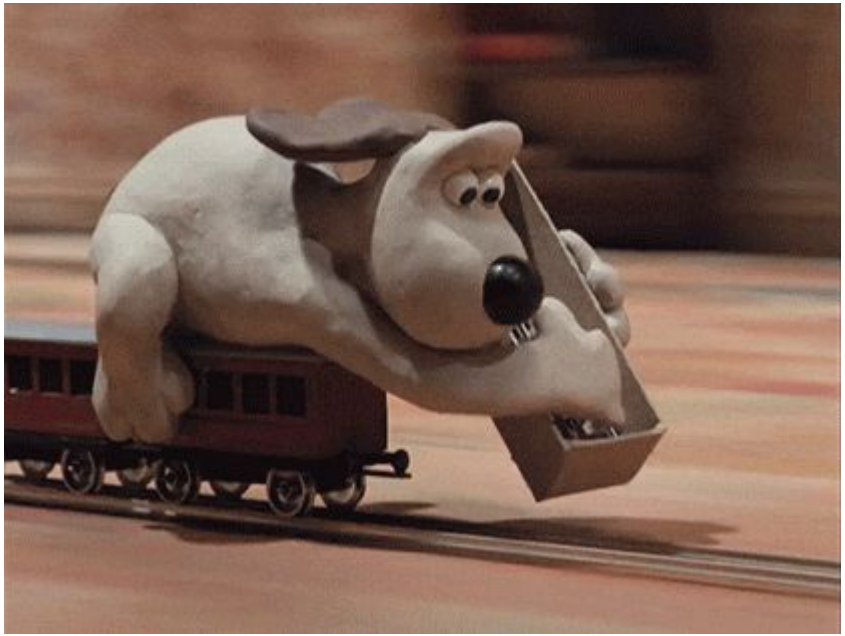
Nathan Marz. Big Data: Principles and best practices of scalable realtime data systems. Simon and Schuster, 2015. Chapter 1: A new paradigm for Big Data

Martin Kleppmann. Designing Data-Intensive Applications. OReilly. 2017.

Learning Goals

- Organize different data management solutions and their tradeoffs
- Understand the scalability challenges involved in large-scale machine learning and specifically deep learning
- Explain the tradeoffs between batch processing and stream processing and the lambda architecture
- Recommend and justify a design and corresponding technologies for a given system

Adding capacity



Stories of catastrophic success?

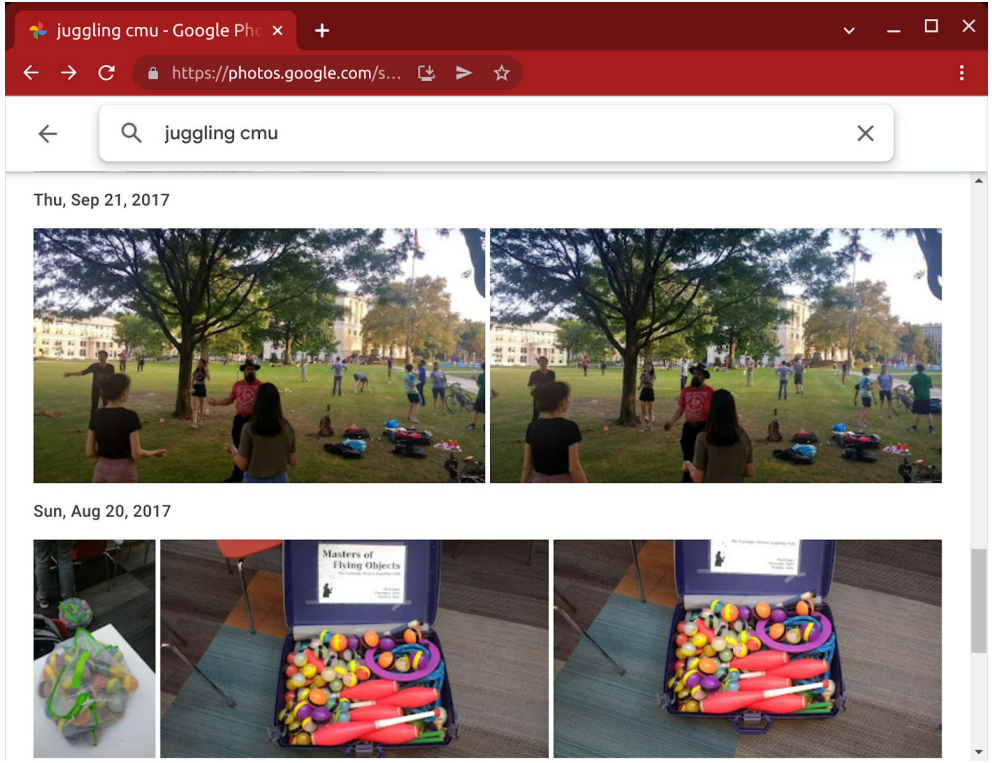
Scaling Strategies

Efficient Algorithms

Faster Machines

More Machines

Case Study


juggling cmu - Google Pho
https://photos.google.com/s...
juggling cmu
Thu, Sep 21, 2017
Sun, Aug 20, 2017
Masters of
Flying Objects

Data Management and Processing in ML-Enabled Systems

Kinds of Data

- Training data
- Input data
- Telemetry data
- (Models)

all potentially with huge total volumes and high throughput

need strategies for storage and processing

Data Management and Processing in ML-Enabled Systems

Store, clean, and update training data

Learning process reads training data, writes model

Prediction task (inference) on demand or precomputed

Individual requests (low/high volume) or large datasets?

Often both learning and inference data heavy, high volume tasks

Distributed Everything

Distributed data cleaning

Distributed feature extraction

Distributed learning

Distributed large prediction tasks

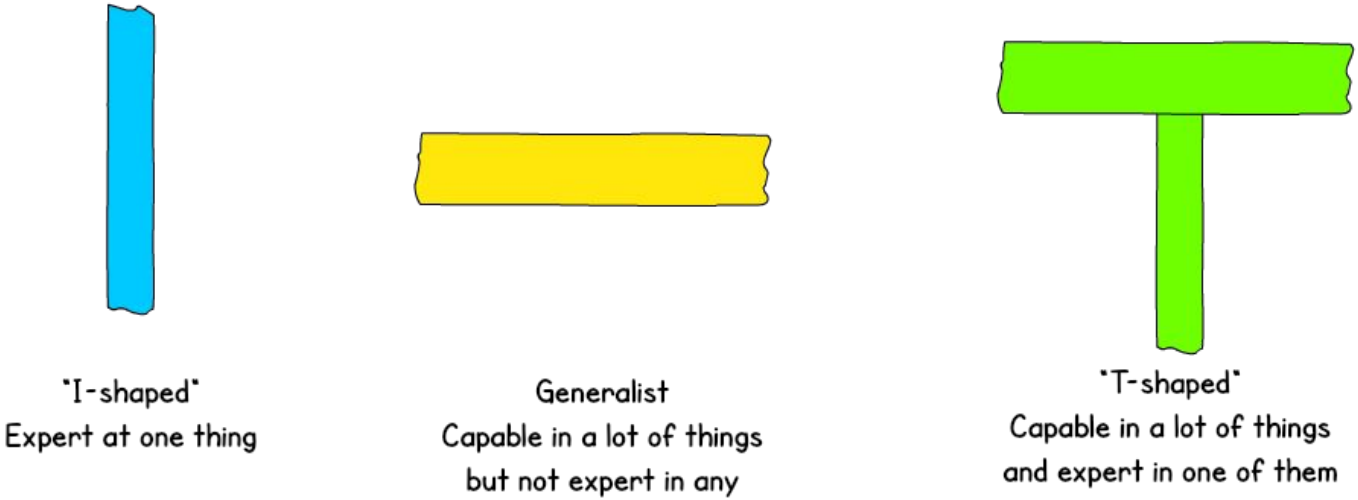
Incremental predictions

Distributed logging and telemetry

Distributed Systems and AI-Powered Systems

- Learning tasks can take substantial resources
- Datasets too large to fit on single machine
- Nontrivial inference time, many many users
- Large amounts of telemetry
- Experimentation at scale
- Models in safety critical parts
- Mobile computing, edge computing, cyber-physical systems

Reminder: T-Shaped People



Go deeper with: Martin Kleppmann. Designing Data-Intensive Applications. OReilly. 2017.

Data Storage Basics

Relational Data Models

Photos:

photo_id	user_id	path	upload_date	size	camera_id	camera_setting
133422131	54351	/st/u211/1U6uFl47Fy.jpg	2021-12-03T09:18:32.124Z	5.7	663	ƒ/1.8; 1/120; 4.44mm; ISO271
133422132	13221	/st/u11b/MFxlL1FY8V.jpg	2021-12-03T09:18:32.129Z	3.1	1844	ƒ/2, 1/15, 3.64mm, ISO1250
133422133	54351	/st/x81/ITzhcSmv9s.jpg	2021-12-03T09:18:32.131Z	4.8	663	ƒ/1.8; 1/120; 4.44mm; ISO48

Users:

user_id	account_name	photos_total	last_login
54351	ckaestne	5124	2021-12-08T12:27:48.497Z
13221	eva.burk	3	2021-12-21T01:51:54.713Z

Cameras:

camera_id	manufacturer	print_name
663	Google	Google Pixel 5
1844	Motorola	Motorola MotoG3

```
select p.photo_id, p.path, u.photos_total
from photos p, users u
where u.user_id=p.user_id and u.account_name = "ckaestne"
```

Document Data Models

```
{
    "_id": 133422131,
    "path": "/st/u211/1U6uFl47Fy.jpg",
    "upload_date": "2021-12-03T09:18:32.124Z",
    "user": {
        "account_name": "ckaestne",
        "account_id": "a/54351"
    },
    "size": "5.7",
    "camera": {
        "manufacturer": "Google",
        "print_name": "Google Pixel 5",
        "settings": "ƒ/1.8; 1/120; 4.44mm; ISO271"
    }
}
```

```
db.getCollection('photos').find( { "user.account_name": "ckaestne"})
```

Log files, unstructured data

```
02:49:12 127.0.0.1 GET /img13.jpg 200
02:49:35 127.0.0.1 GET /img27.jpg 200
03:52:36 127.0.0.1 GET /main.css 200
04:17:03 127.0.0.1 GET /img13.jpg 200
05:04:54 127.0.0.1 GET /img34.jpg 200
05:38:07 127.0.0.1 GET /img27.jpg 200
05:44:24 127.0.0.1 GET /img13.jpg 200
06:08:19 127.0.0.1 GET /img13.jpg 200
```

Tradeoffs

Data Encoding

Plain text (csv, logs)

Semi-structured, schema-free (JSON, XML)

Schema-based encoding (relational, Avro, ...)

Compact encodings (protobuffer, ...)

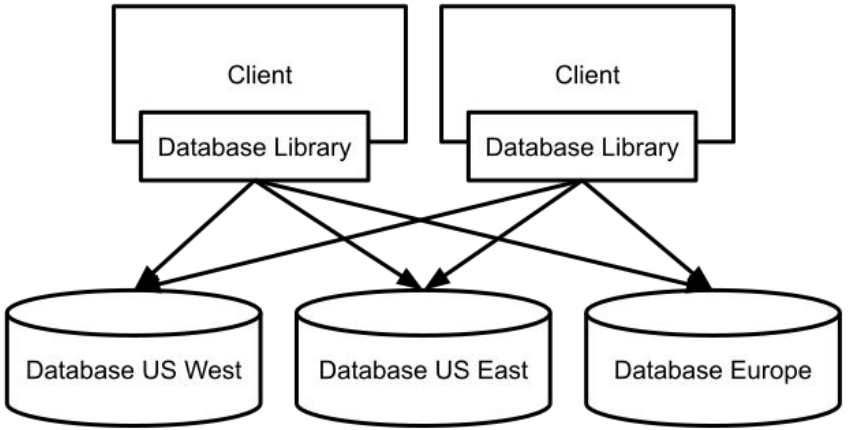
Distributed Data Storage

Replication vs Partitioning

Partitioning

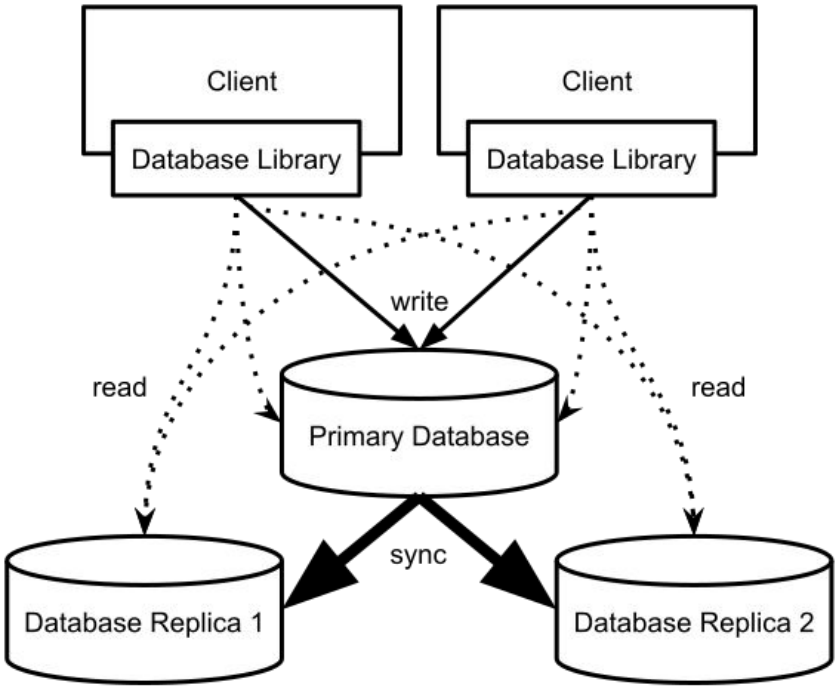
Divide data:

- *Horizontal partitioning:* Different rows in different tables; e.g., movies by decade, hashing often used
- *Vertical partitioning:* Different columns in different tables; e.g., movie title vs. all actors



Tradeoffs?

Replication with Leaders and Followers



Replication Strategies: Leaders and Followers

Write to leader, propagated synchronously or async.

Read from any follower

Elect new leader on leader outage; catchup on follower outage

Built in model of many databases (MySQL, MongoDB, ...)

Benefits and Drawbacks?

Example: Google File System

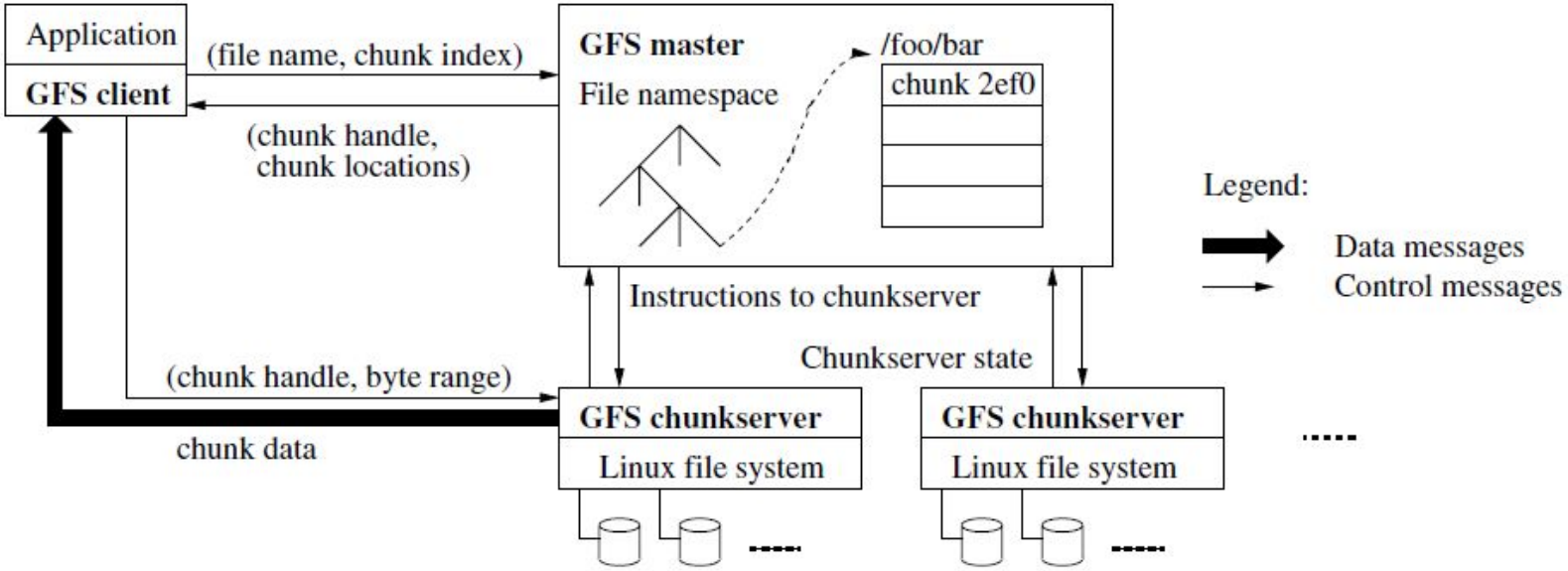


Figure 1: GFS Architecture

Beyond Leaders and Followers

Multi-Leader Replication
- Scale write access, add redundancy
- Requires coordination among leaders and resolution of write conflicts
- Offline leaders (e.g. apps), collaborative editing (like Google Docs), *Postgres BDR*, …

Leaderless Replication
- Client writes to multiple replica, propagate from there
- Versioning of entries (clock problem)
- Read from multiple replica (quorum required), version repair on reads/background repair process
- e.g. *Amazon Dynamo*, *Cassandra*, *Voldemort*

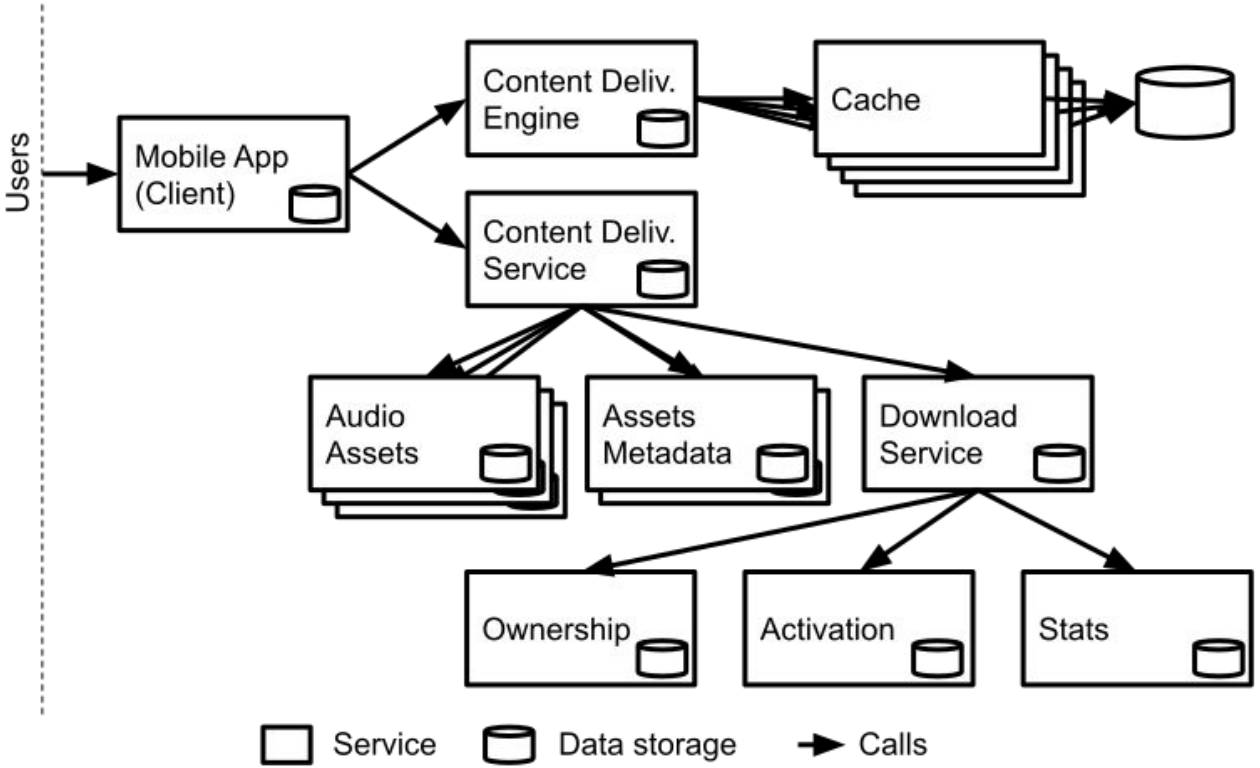
Transactions
- Multiple operations conducted as one, all or nothing
- Avoids problems such as dirty reads, dirty writes
- Various strategies, including locking and optimistic+rollback
- Overhead in distributed setting

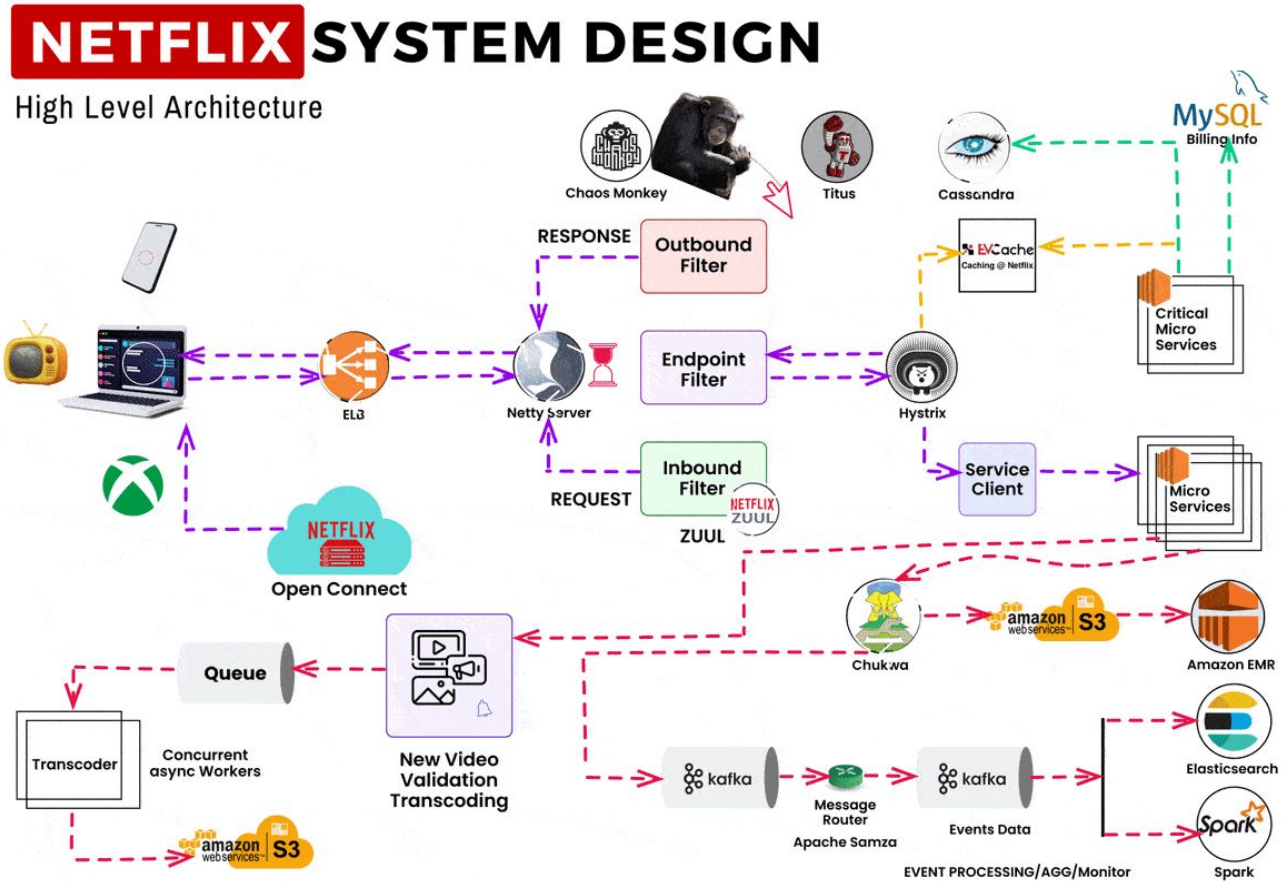
Data Processing (Overview)

- Services (online)
  - Responding to client requests as they come in
  - Evaluate: Response time
- Batch processing (offline)
  - Computations run on large amounts of data
  - Takes minutes to days; typically scheduled periodically
  - Evaluate: Throughput
- Stream processing (near real time)
  - Processes input events, not responding to requests
  - Shortly after events are issued

Microservices

Microservices




NETFLIX SYSTEM DESIGN
High Level Architecture
Chaos Monkey
Titus
Cassandra
MySQL
Billing Info
RESPONSE
Outbound Filter
EVCache
Caching @ Netflix
Critical Micro Services
ELB
Netty Server
Endpoint Filter
Hystrix
Inbound Filter
REQUEST
ZUUL
Service Client
Micro Services
NETFLIX
Open Connect
Chukwa
amazon web services S3
Amazon EMR
Queue
Transcoder
Concurrent async Workers
New Video Validation Transcoding
kafka
Message Router
Apache Samza
kafka
Events Data
Elasticsearch
Spark
EVENT PROCESSING/AGG/Monitor
amazon web services S3

Microservices

Independent, cohesive services

- Each specialized for one task
- Each with own data storage
- Each independently scalable through multiple instances + load balancer

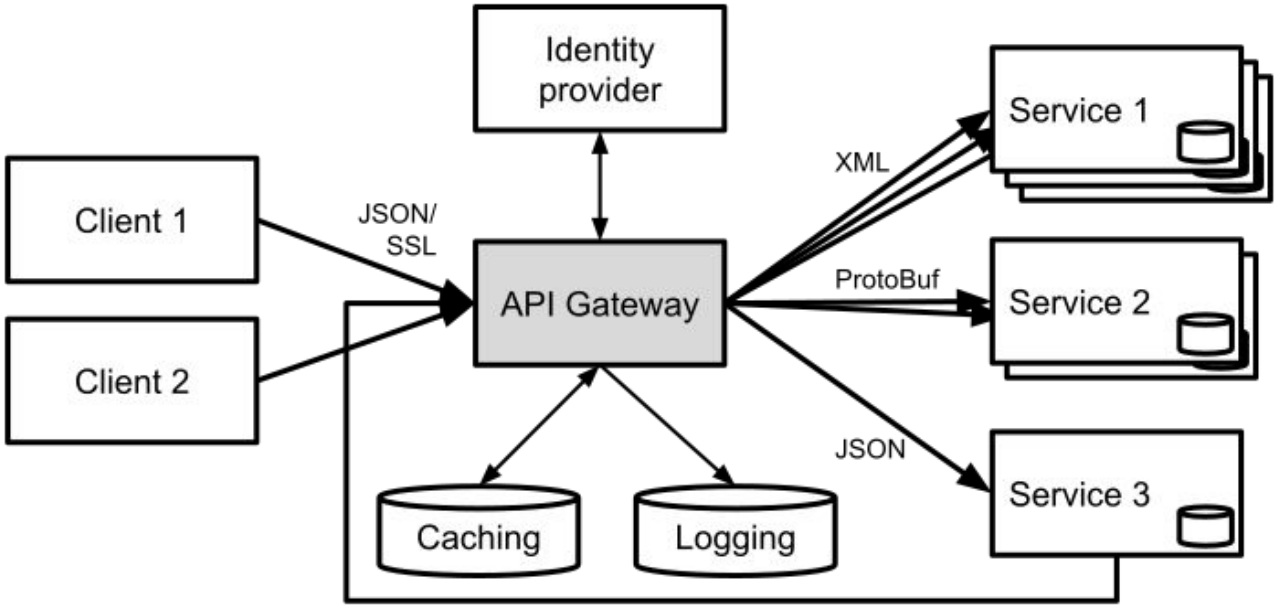
Remote procedure calls

Different teams can work on different services independently (even in different languages)

But: Substantial complexity from distributed system nature: various network failures, latency from remote calls, ...

Avoid microservice complexity unless really needed for scalability

API Gateway Pattern



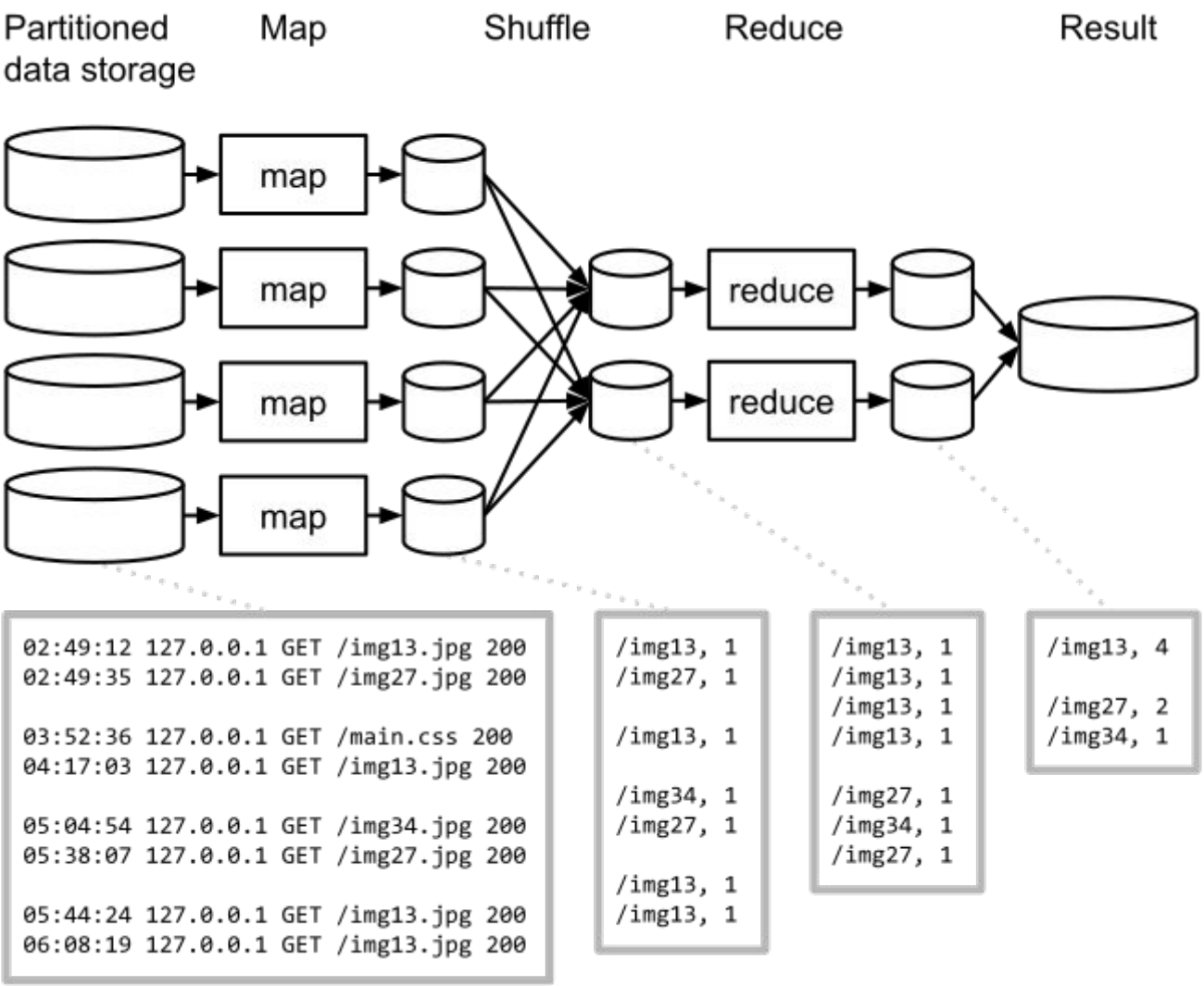
Central entry point, authentication, routing, updates, ...

Batch Processing

Large Jobs

```
cat /var/log/nginx/access.log |
    awk '{print $7}' |
    sort |
    uniq -c |
    sort -r -n |
    head -n 5
```

- Analyzing TB of data, typically distributed storage
- Filtering, sorting, aggregating
- Producing reports, models, ...


Partitioned data storage
Map
Shuffle
Reduce
Result
map
map
map
map
reduce
reduce
02:49:12 127.0.0.1 GET /img13.jpg 200
02:49:35 127.0.0.1 GET /img27.jpg 200
03:52:36 127.0.0.1 GET /main.css 200
04:17:03 127.0.0.1 GET /img13.jpg 200
05:04:54 127.0.0.1 GET /img34.jpg 200
05:38:07 127.0.0.1 GET /img27.jpg 200
05:44:24 127.0.0.1 GET /img13.jpg 200
06:08:19 127.0.0.1 GET /img13.jpg 200
/img13, 1
/img27, 1
/img13, 1
/img34, 1
/img27, 1
/img13, 1
/img13, 1
/img13, 1
/img13, 1
/img13, 1
/img34, 1
/img27, 1
/img13, 4
/img27, 2
/img34, 1

Distributed Batch Processing

Process data locally at storage

Aggregate results as needed

Separate pluming from job logic

MapReduce as common framework

MapReduce -- Functional Programming Style

Similar to shell commands: Immutable inputs, new outputs, avoid side effects

Jobs can be repeated (e.g., on crashes)

Easy rollback

Multiple jobs in parallel (e.g., experimentation)

Machine Learning and MapReduce

Dataflow Engines (Spark, Tez, Flink, ...)

Single job, rather than subjobs

More flexible than just map and reduce

Multiple stages with explicit dataflow between them

Often in-memory data

Plumbing and distribution logic separated

Key Design Principle: Data Locality

> *Moving Computation is Cheaper than Moving Data*
>
> -- *Hadoop Documentation*

Data often large and distributed, code small

Avoid transferring large amounts of data

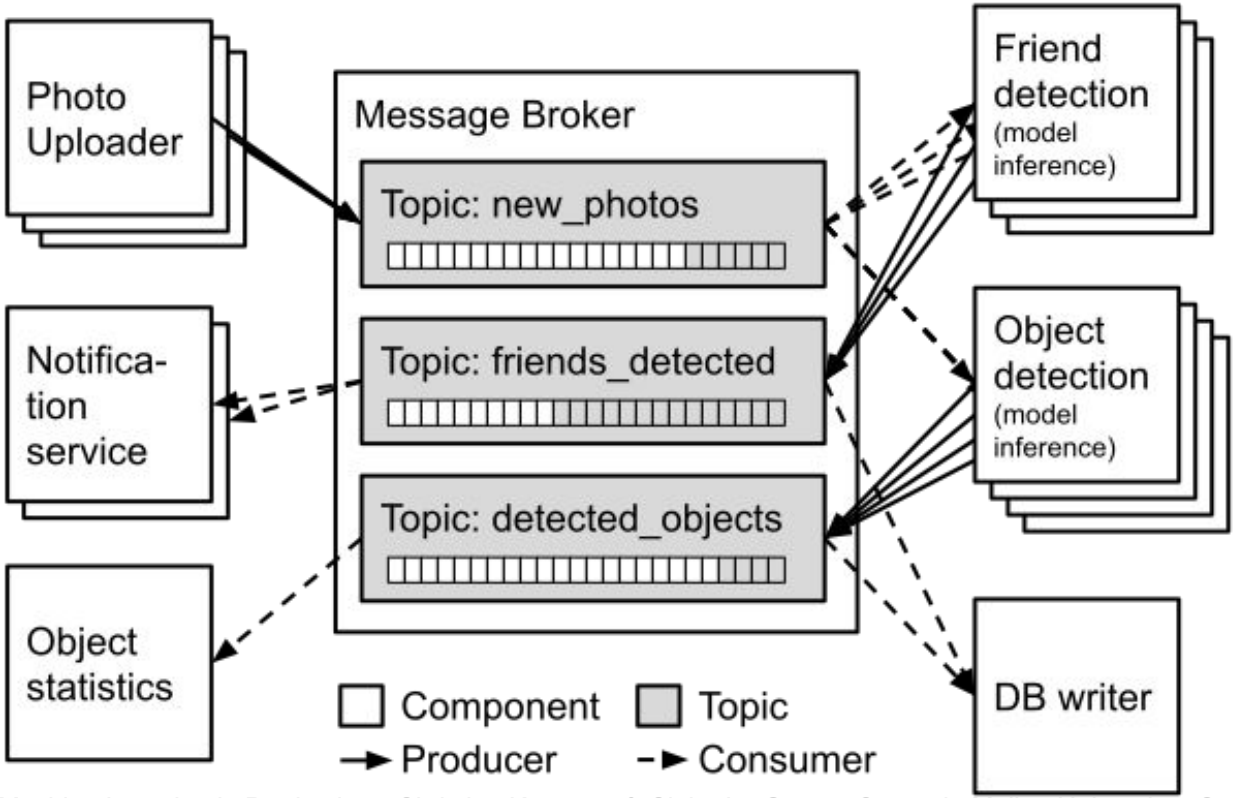
Perform computation where data is stored (distributed)

Transfer only results as needed

"The map reduce way"

Stream Processing

Stream Processing (e.g., Kafka)



Messaging Systems

Multiple producers send messages to topic

Multiple consumers can read messages

-> Decoupling of producers and consumers

Message buffering if producers faster than consumers

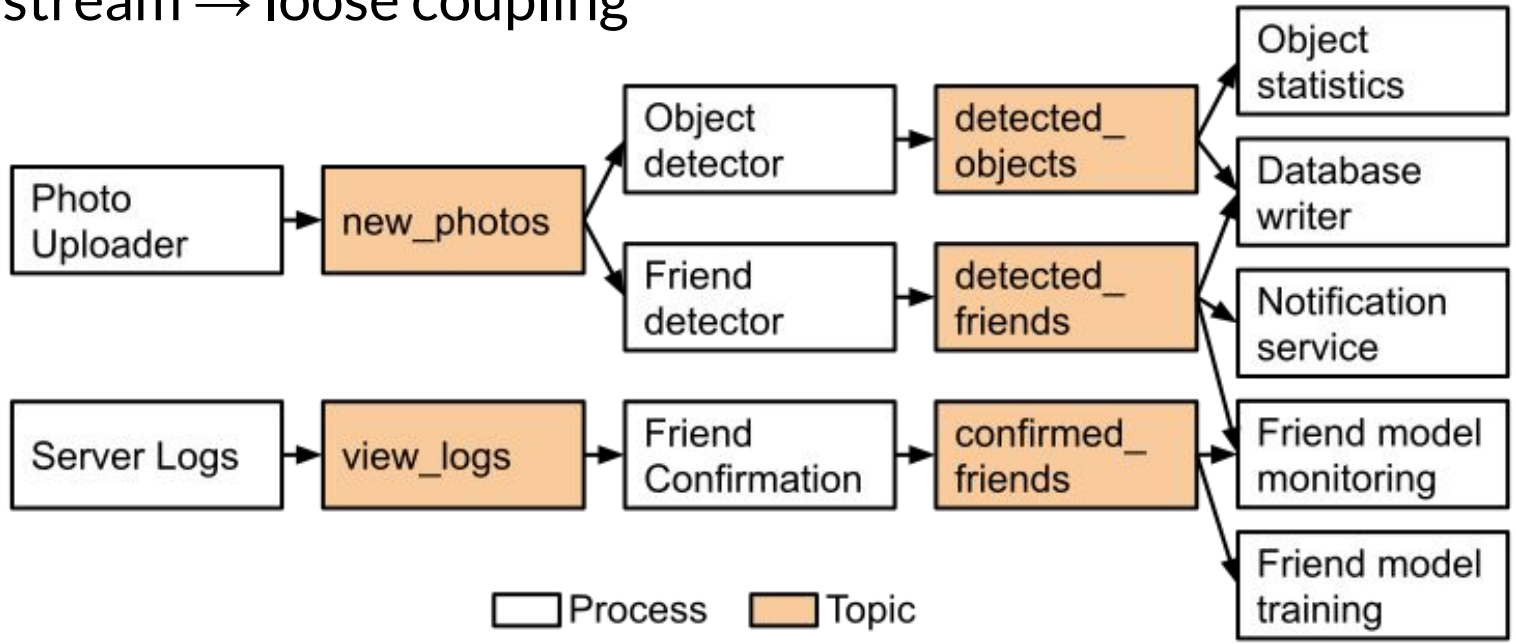
Typically some persistency to recover from failures

Messages removed after consumption or after timeout

Various error handling strategies (acknowledgements, redelivery, ...)

Common Designs

Like shell programs: Read from stream, produce output in other stream → loose coupling



Stream Queries

Processing one event at a time independently

vs incremental analysis over all messages up to that point

vs floating window analysis across recent messages

Works well with probabilistic analyses

Consumers

Multiple consumers share topic for scaling and load balancing

Multiple consumers read same message for different work

Partitioning possible

Design Questions

Message loss important? (at-least-once processing)

Can messages be processed repeatedly (at-most-once processing)

Is the message order important?

Are messages still needed after they are consumed?

Stream Processing and AI-enabled Systems?

Reasoning about Dataflows

Many data sources, many outputs, many copies

Which data is derived from what other data and how?

Is it reproducible? Are old versions archived?

How do you get the right data to the right place in the right format?

Plan and document data flows

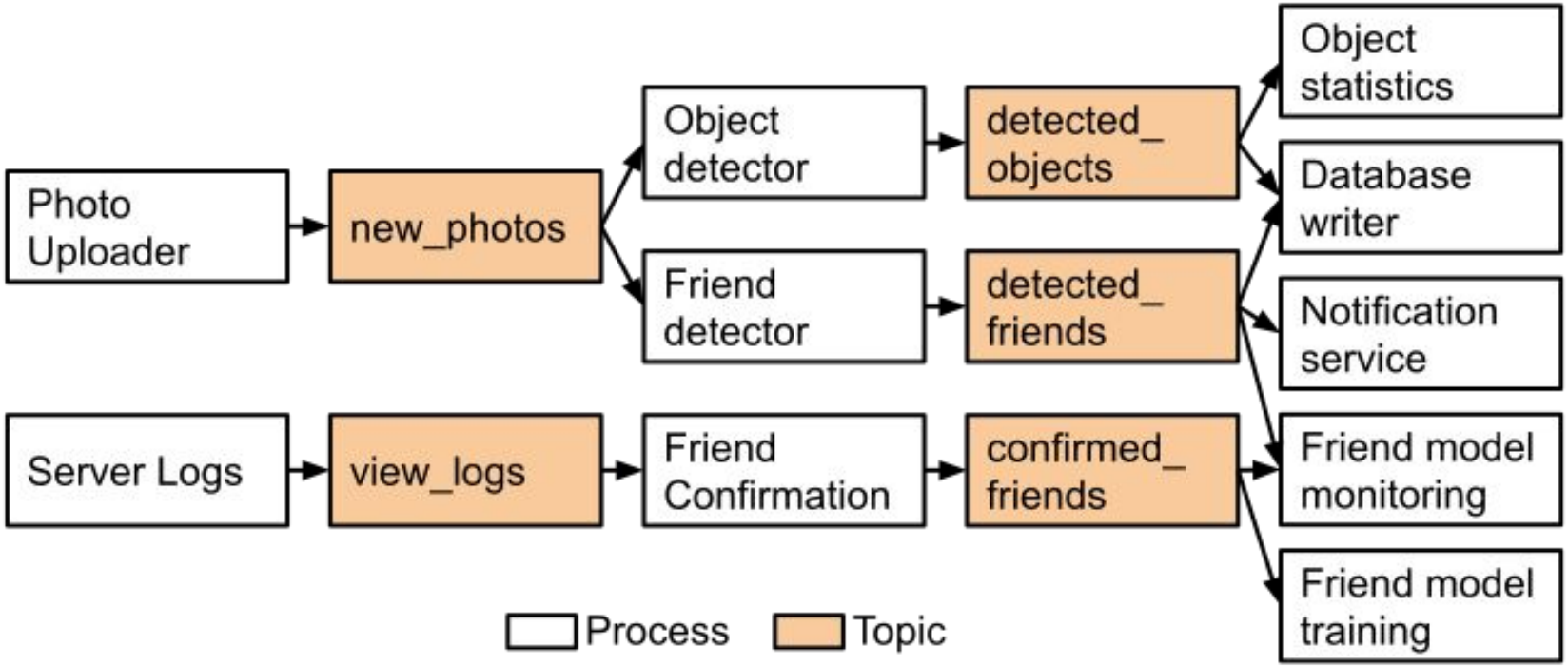


Photo Uploader
new_photos
Object detector
Friend detector
detected_ objects
detected_ friends
Object statistics
Database writer
Notification service
Server Logs
view_logs
Friend Confirmation
confirmed_ friends
Friend model monitoring
Friend model training
Process
Topic

Event Sourcing

- Append only databases
- Record edit events, never mutate data
- Compute current state from all past events, can reconstruct old state
- For efficiency, take state snapshots
- *Similar to traditional database logs, but persistent*

```
addPhoto(id=133422131, user=54351, path="/st/u211/1U6uFl47Fy.jpg", date="2021-12-03T09:18:32.124Z")
updatePhotoData(id=133422131, user=54351, title="Sunset")
replacePhoto(id=133422131, user=54351, path="/st/x594/vipxBMFlLF.jpg", operation="/filter/palma")
deletePhoto(id=133422131, user=54351)
```

Benefits of Immutability (Event Sourcing)

- All history is stored, recoverable
- Versioning easy by storing id of latest record
- Can compute multiple views
- Compare *git*

  On a shopping website, a customer may add an item to their cart and then remove it again. Although the second event cancels out the first event [...], it may be useful to know for analytics purposes that the customer was considering a particular item but then decided against it. Perhaps they will choose to buy it in the future, or perhaps they found a substitute. This information is recorded in an event log, but would be lost in a database [...].

Drawbacks of Immutable Data

The Lambda Architecture

3 Layer Storage Architecture

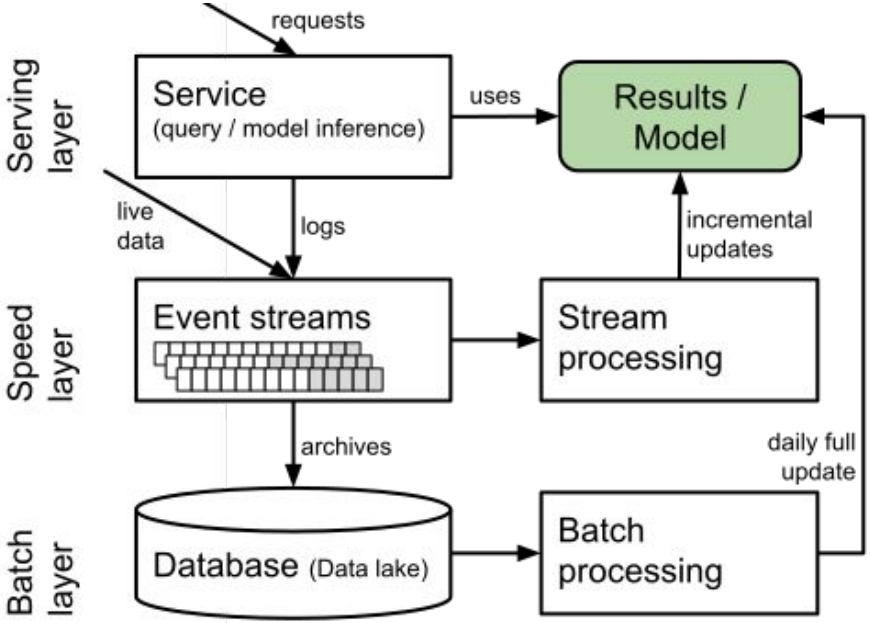
- Batch layer: best accuracy, all data, recompute periodically
- Speed layer: stream processing, incremental updates, possibly approximated
- Serving layer: provide results of batch and speed layers to clients

Assumes append-only data

Supports tasks with widely varying latency

Balance latency, throughput and fault tolerance

Lambda Architecture and Machine Learning



- Learn accurate model in batch job
- Learn incremental model in stream processor

Data Lake

Trend to store all events in raw form (no consistent schema)

May be useful later

Data storage is comparably cheap

Data Lake

Trend to store all events in raw form (no consistent schema)

May be useful later

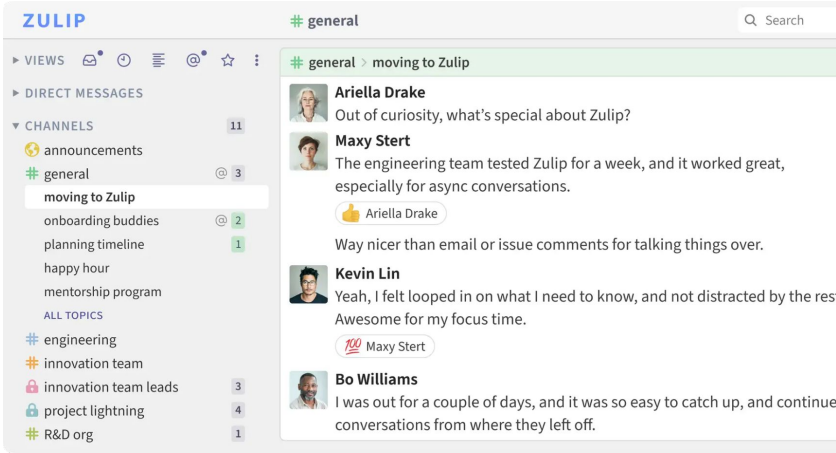
Data storage is comparably cheap

Bet: *Yet unknown future value of data is greater than storage costs*

Breakout: Scaling Zulip

Scenario: Scaling Zulip for very large company (>50k users).

In addition to summarization (I1), need ML-based anomaly detection to compromised accounts (insider treats).



As a group, discuss and post in `#lecture`, tagging group members:

- How to distribute storage: [partitioning/replication]
- How to design scalable summarization for user interface: [service/stream/batch/lambda]
- How to design scalable training of anomaly detection model: [service/stream/batch/lambda]

Excursion: ETL Tools

ETL: Extract, Transform, Load

- Transfer data between data sources, often OLTP -> OLAP system
- Many tools and pipelines
  - Extract data from multiple sources (logs, JSON, databases), snapshotting
  - Transform: cleaning, (de)normalization, transcoding, sorting, joining
  - Loading in batches into database, staging
- Automation, parallelization, reporting, data quality checking, monitoring, profiling, recovery
- Many commercial tools

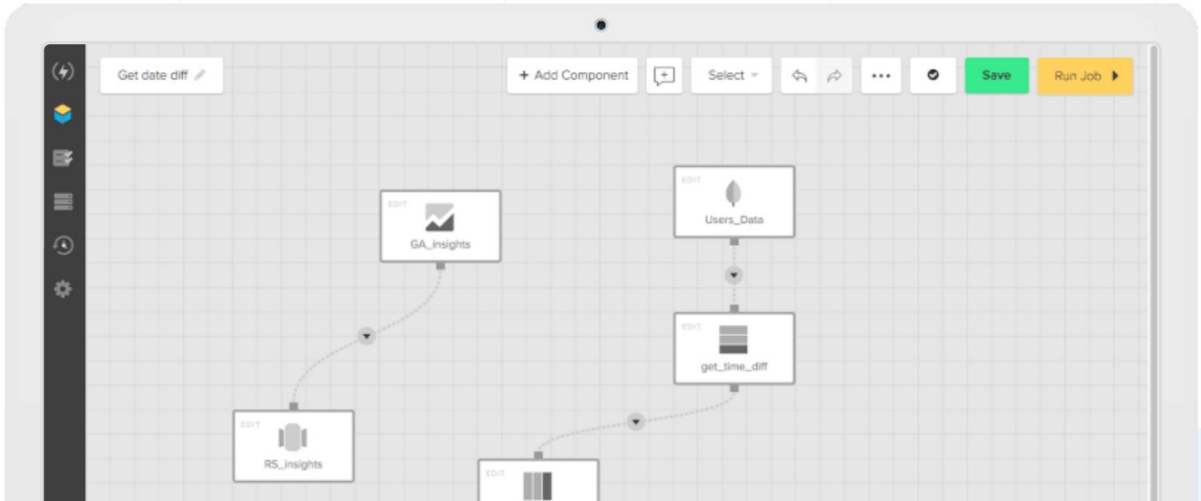
Examples of tools in several lists

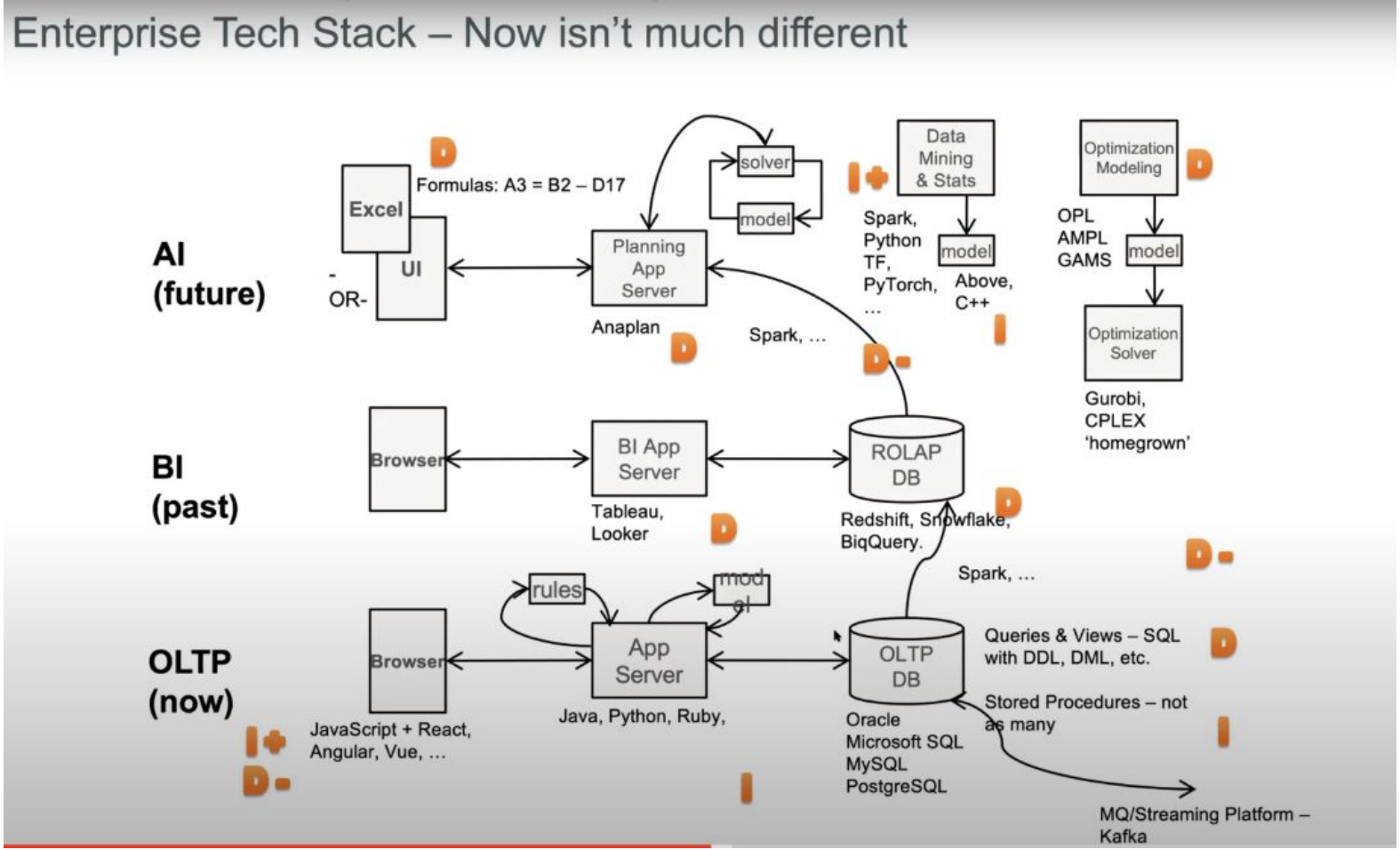
Xplenty

The leading data integration platform to bring all your data sources together.

Create simple, visualized data pipelines to your data warehouse or data lake.

GET STARTED





Molham Aref "[Business Systems with Machine Learning]"

Excursion:
Performance Planning and Analysis

Performance Planning and Analysis

Ideally architectural planning upfront

- Identify key components and their interactions
- Estimate performance parameters
- Simulate system behavior (e.g., queuing theory)

Existing system: Analyze performance bottlenecks

- Profiling of individual components
- Performance testing (stress testing, load testing, etc)
- Performance monitoring of distributed systems

Performance Testing

Load testing: Assure handling of maximum expected load

Scalability testing: Test with increasing load

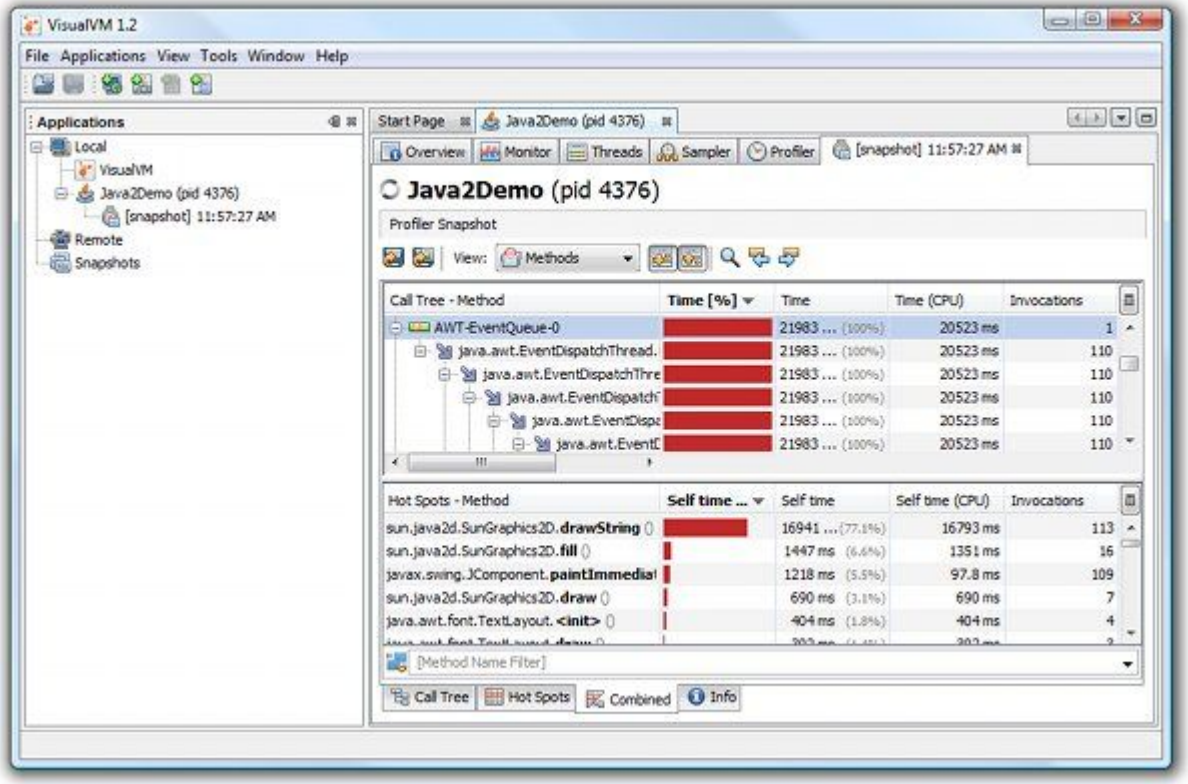
Soak/spike testing: Overload application for some time, observe stability

Stress testing: Overwhelm system resources, test graceful failure + recovery

Observe (1) latency, (2) throughput, (3) resource use

All automateable; tools like JMeter

Profiling


VisualVM 1.2
File Applications View Tools Window Help
Applications
Local
VisualVM
Java2Demo (pid 4376)
[snapshot] 11:57:27 AM
Remote
Snapshots
Start Page
Java2Demo (pid 4376)
Overview
Monitor
Threads
Sampler
Profiler
[snapshot] 11:57:27 AM
Java2Demo (pid 4376)
Profiler Snapshot
View:
Methods
Call Tree - Method
Time [%]
Time
Time (CPU)
Invocations
AWT-EventQueue-0
java.awt.EventDispatchThread.
java.awt.EventDispatchThre
java.awt.EventDispatch
java.awt.EventDispa
java.awt.EventD
21983 ... (100%)
20523 ms
1
110
110
110
110
Hot Spots - Method
Self time ...
Self time
Self time (CPU)
Invocations
sun.java2d.SunGraphics2D.drawString ()
sun.java2d.SunGraphics2D.fill ()
javax.swing.JComponent.paintImmedia
sun.java2d.SunGraphics2D.draw ()
java.awt.font.TextLayout.<init> ()
16941 ... (77.1%)
16793 ms
113
1447 ms (6.6%)
1351 ms
16
1218 ms (5.5%)
97.8 ms
109
690 ms (3.1%)
690 ms
7
404 ms (1.8%)
404 ms
4
[Method Name Filter]
Call Tree
Hot Spots
Combined
Info

Performance Monitoring of Distr. Systems

Source: https://blog.appdynamics.com/tag/fiserv/

Performance Monitoring of Distributed Systems

Instrumentation of (Service) APIs

Load of various servers

Typically measures: latency, traffic, errors, saturation

Monitoring long-term trends

Alerting

Automated releases/rollbacks

Canary testing and A/B testing

Summary

Large amounts of data (training, inference, telemetry, models)

Distributed storage and computation for scalability

Common design patterns (e.g., batch processing, stream processing, lambda architecture)

Design considerations: mutable vs immutable data

Distributed computing also in machine learning; many challenges through distribution: failures, debugging, performance, …

Lots of tooling for data extraction, transformation, processing

Recommended reading: Martin Kleppmann. Designing Data-Intensive Applications. OReilly. 2017.

Even More Recommended Readings

- Molham Aref "Business Systems with Machine Learning" Invited Talk 2020
- Sawadogo, Pegdwendé, and Jérôme Darmont. "On data lake architectures and metadata management." Journal of Intelligent Information Systems 56, no. 1 (2021): 97-120.
- Warren, James, and Nathan Marz. Big Data: Principles and best practices of scalable realtime data systems. Manning, 2015.
- Smith, Jeffrey. Machine Learning Systems: Designs that Scale. Manning, 2018.
- Polyzotis, Neoklis, Sudip Roy, Steven Euijong Whang, and Martin Zinkevich. 2017. “Data Management Challenges in Production Machine Learning.” In Proceedings of the 2017 ACM International Conference on Management of Data, 1723–26. ACM.